Instructions for exam

Hello!

If a question is unclear, please write down your assumptions about the question when you are answering it.

The exam has a total of 40 points and you need 21 points to pass the exam. The grade is on the scale U/G.

Answers must be written in English. Dictionaries may be used.

It is possible to get partial credit for the questions that aren't multiple choice.

A checklist of common mistakes that cost points:

- Answer all questions. A bad answer never gives less points than no answer.
- Read the question again after you have written the answer. Verify that you have actually answered the question. Verify that you answered all parts. Verify that you have not hidden the answer between many other irrelevant comments about the topic. *Cursives are added to highlight the key elements of the questions.*
- In particular, don't forget to give an example if that is requested.
- When a question asks you to compare two things A and B, make sure to highlight the contrasts: their differences. I do not want a full description of A and a full description of B, leaving it to me to find the differences. It is better to use tables than paragraphs of text to compare.

I likely will not be able to make it to the exam hall today due to scheduling conflicts. All the best today and I hope this marks the beginning of a lovely summer.

1 Multiple Choice

Each question in the multiple choice section is worth 1 point. There is only one expected best answer for each question.

What is the primary goal of risk management in software engineering?

- To remove all possible risks from the project
- To allocate extra budget for unexpected failures
- To ensure the software is developed without any defects
- To identify, assess, and mitigate potential risks

Which technique is used to gather requirements from stakeholders?

- Code inspection
- Continuous deployment
- Requirements elicitation
- Debugging

Which of the following software development models follows a linear, sequential approach?

- Waterfall
- Agile
- Incremental
- Spiral

What is the main difference between black-box testing and glass-box testing?

- Black-box testing focuses on internal code structure, while glass-box testing examines system functionality
- Black-box testing does not require access to the code, while glass-box testing requires knowledge of the code structure
- Black-box testing only applies to unit tests, while glass-box testing applies to integration tests
- Black-box testing uses source code analysis, while glass-box testing tests from a user perspective

How does Test-Driven Development (TDD) differ from traditional development approaches?

- TDD writes tests after implementing the functionality
- TDD requires writing tests before writing the actual code
- TDD eliminates the need for validation and verification
- TDD focuses only on UI testing

Which of the following best describes continuous integration?

- A version control approach that delays integration until the end of a project
- A method of organizing teams in Agile development
- A development practice where code changes are frequently integrated into the main branch and tested
- A testing method used for finding performance issues

In peer review, how does the process typically work?

- The entire team meets in a formal session to discuss previously reviewed code
- The reviewer inspects a portion of source code and provides feedback
- The reviewer makes modifications directly to the programmer’s code
- One programmer presents their work while another follows along in real time

What is the primary purpose of code reviews?

- To reduce defect rates and improve maintainability
- To allow managers to evaluate programmers' performance
- To completely eliminate all bugs before release
- To rewrite code from scratch

Which of the following is a key element of Agile planning?

- One-time budgeting at the start of the project
- Fixed project scope
- Iteration planning and release planning
- Long-term fixed milestones

In project planning, what does a Gantt Chart (Activity bar chart) primarily help visualize?

- The project schedule and task dependencies
- The relationship between stakeholders
- The project's overall quality
- The functional and non-functional requirements

What is one key benefit of having an explicit architecture?

- Ensures the system design never needs modifications
- Helps analyze whether the system meets non-functional requirements
- Eliminates the need for software documentation
- Reduces the need for stakeholder communication

What is a challenge with requirements elicitation?

- There is no need to validate requirements after they are collected
- Requirements remain constant throughout the development process
- Stakeholders may not fully understand or articulate their needs
- Stakeholders always agree on requirements

What is the COCOMO model used for?

- Evaluating software security risks
- Creating Agile sprint backlogs
- Managing team roles in software projects
- Estimating the cost of software development

Maximum marks: 13

2 Requirements

What is a difference between user requirements and system requirements (1pt)?

Fill in your answer here

A local pharmacy has decided to start an online platform. They have hired your company to create the software system.

Identify three stakeholders and an example of a functional requirement they would be interested in. (3 pt)

Fill in your answer here

The client has stated the following, "I want the customers to be able to find their medication easily."

From this statement, write an example of the following **two** things. You are allowed to make assumptions if you don't have enough information.

(a) - a user story (1 pt)

Fill in your answer here

(b) - a non-functional requirement (1 pt)

Fill in your answer here

Maximum marks: 6

3 Validation and Verification

In the context of software engineering, differentiate between validation and verification. (2 pt)

Fill in your answer here

Describe how validation can be assessed by explaining two techniques that can be used to ensure requirements correctness. (2 pt)

Fill in your answer here

Maximum marks: 4

4 Code Inspection

A company is debating between using **pair programming** and **pre-checking** for their code inspection process. Name three factors should they consider when choosing between these two approaches? (3 pts)

Fill in your answer here

Maximum marks: 3

5 Architecture

Consider a hospital management system that uses an **MVC architecture**. The management system tracks things such as inventory, patients, employee and patient information. Identify the likely responsibilities of the **Model, View, and Controller** components in a possible use case of the software. This use case does not need to be complex. A sketch may be useful. (5 pt)

Fill in your answer here

Format

Words: 0

Maximum marks: 5

6 Project Management Diagrams

Here is an activity table:

Task	Effort (person-days)	Duration (days)	Dependencies
T1	6	3	
T2	4	4	T1 (M1)
T3	1	1	T1 (M1)
T4	4	8	T3 (M2)

Sketch a Gantt chart (3 pt). What are two pieces of information you would need in order to create a staff allocation chart (2 pt)?

Fill in your answer here

Format

Words: 0

Maximum marks: 5

7 Legacy Systems

A legacy system in a hospital is still in use but has no formal documentation. What are two metrics that can be used to assess whether it should be maintained or replaced? A diagram may be useful. Explain your answer. (4 pt)

Fill in your answer here

Format

Words: 0

Maximum marks: 4